

Ronak Varshney

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EDUCATION

Aligarh College Of Engineering And Technology

Bachelor of Technology in Computer Science and Engineering, 8.42 CGPA

Aligarh, Uttar Pradesh

Sep. 2022 – July 2026

Dharam Samaj Bal Mandir

Intermediate PCM 92.3 percent

Aligarh, Uttar Pradesh

Aug. 2021 – June 2022

EXPERIENCE

Software Development Engineer Intern - The Squirrel (Remote) | July 2025 – Present

- Architected and delivered **3+ production-grade MERN applications**, supporting **5K–10K+ monthly active users**, improving workflow automation efficiency by **~40%** through AI-driven features.
- Designed and optimized **RESTful APIs (~50+ endpoints)**, reducing average response latency by **38%** via query optimization, indexing, and caching strategies. Implemented **Redis caching layer and session management**, cutting database read load by **~45%** and improving high-traffic endpoint performance
- Built **asynchronous job processing pipelines using BullMQ**, handling **10K+ background jobs/day** (notifications, scheduling, data pipelines) with improved reliability and fault tolerance.
- Led frontend performance optimizations by converting Figma designs into **highly responsive React components**, improving **LCP by ~25%** and reducing UI render delays. Collaborated in an agile team to deliver **multi-client (including international) projects**, ensuring scalable architecture, clean code practices, and maintainability.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (Postgres, MySQL), JavaScript, HTML/CSS, TypeScript

Backend Frameworks: Node.js, Express.js, Auth.js, Jwt, RestAPIs

Frontend Frameworks: React, Next.js, Framer Motion, Tailwind CSS, Zustand, Expo, Context API

AI/LLM Development: Langchain, LangGraph, RAG pipelines

Database: MongoDB, PostgreSQL, MySQL, Supabase, Prisma ORM

Developer Tools And AI Tools: Git, Docker, Linux, Visual Studio, Vercel, Claude, Lovable, Gemini, Chatgpt

PROJECTS

Code Canvas | Next.js, TypeScript, Socket.io, Docker

- Engineered a **WebSocket-based real-time synchronization engine**, enabling **many concurrent users per room** with low-latency.
- Designed **room-based architecture with event batching and throttling**, reducing network overhead by **~30%** during peak collaboration.
- Implemented **multi-language code execution and live chat**, improving user engagement and session duration.
- Containerized services using Docker, ensuring **consistent deployments and scalability across environments**. [Demo](#)

MindEase | MERN, Next.js, Langchain, LangGraph, Socket.io, Docker, Framer Motion

- Built an AI-driven conversational system using **RAG architecture**, improving response relevance by **~35%** through optimized retrieval and context handling
- Developed a **personalized mental wellness assessment engine**, increasing user retention by **~25%** via adaptive recommendations.
- Designed secure authentication and data pipelines, ensuring **100% protected user sessions** with JWT-based auth and validation layers.
- Implemented real-time chat system supporting **concurrent users with stable low-latency messaging**. Integrated **donation and outreach modules**, improving accessibility and engagement for underprivileged users. [Demo](#)

Campus Connect (3rd Place - ACET InnovateX) | MERN, Socket.io, Gemini API, Multer, Resend

- Delivered a **full-stack platform in 72 hours**, supporting **3 user roles (Student/Teacher/Admin)** with role-based access control.
- Built **real-time messaging and notification system**, reducing communication delays by **~60%** compared to traditional methods.
- Automated test creation and evaluation workflows, cutting manual effort for instructors by **~50%**.
- Designed analytics dashboards for performance tracking, enabling data-driven insights for academic management. Integrated AI-based content generation, reducing manual notice/event creation time by **~70%**.

ACHIEVEMENTS & CERTIFICATES

Hackathons: 3rd Place at ACET InnovateX (15+ teams) / Certificate of Excellence at AuraFlix (Top 4/400+ participants) — **Open Source:** HacktoberFest 2025, GSOC- 24/25 — **DSA:** Solved 500+ problems on LeetCode & GeeksforGeeks — **Linkedin Autonomous Agent Development certificate.**